

Research Assignment

① Business Analytics

→ This is the process where evaluating business demands and examining solutions to the challenges of the business. Business Analysis is used in business to understand how to meet their short term and long term business goals.

Real world Example

→ Woolworth uses business analytics to analyze sales data across different regions and season, it may identify that demand for certain fresh products increases during holidays allowing the company to adjust inventory and pricing strategies to maximize revenue and reduce waste.

② Data Analytics

→ Data Analytics is when ~~data~~ Raw data is collected and checked to find patterns and useful insights. Data Analytics is focused on "WHAT happened" "Why" did it happen and finding ways to the problem.

Real world Example

→ Busmed hospital analyzes patients data to identify which treatment lead to faster recovery times.

③ Business Intelligence (BI)

→ This is the practice of turning data into actionable insights that enable business leaders to improve the performance of the business this include collecting data, creating models, analyzing the data with queries, create data visualization such as charts and produce reports to be used by business decision makers.

Real world Example

→ We were using google locker and Power BI to create dashboards that show how a specific retail company's monthly sales performance.

④ Data-Driven Decision Making

- This is a management approach where decisions are based on data analysis rather than guesswork or intuition. By leveraging data sources such as customer feedback, market trends, organization can align their strategies with business goals, reduce uncertainty and improve confidence in their decision.

Real-world Example

- ~~Amazon~~ Walmart as an online store decides which ads to run based on click-through rates instead of personal opinion.

⑤ Big Data

- This refers to massive, complex data sets that traditional data management systems cannot handle. When big data is properly collected, managed and analyzed it can help organization discover new insights and make better business decision.

Real-world Example

- Social media Platforms like X generate million of posts and interaction every second contributing to the vast volume of data generated daily.

⑥ Descriptive Analytics

- This is the foundation for further analysis it sets the stage for understanding past performance and current trends. It uses Statistical summaries and data visualization technique to condense and describe historical data.

Real world Example

- A hospital analyze past patients records to calculate average wait time and identify the most common illnesses. This help them understand pattern in patients visits such as busy period or common conditions but it does not explain causes.

① Diagnostic Analytics

- This is the process of using data to determine the cause of trends and correlations between variables. It goes beyond describing results by identifying problem, relationship not just cause but explain why something happened.

Real world Example

- Shoprite analyse sales data and discover that a decline in sales was caused by stock shortage in key products.

② Predictive Analytics

- Predictive Analytics examine past data uncover patterns and relationships and use them as insight to forecast what is likely to happen next.

Real world Example

- calculating the probability of a share rising before investing like at investment.

③ Prescriptive Analytics

- This is a type of data analytic that seeks to understand what is needed to achieve an aim. This requires use data analytic along with technology to enhance business decision.

Real world Example

- A logistic company use algorithm to determine the most efficient delivery routes

④ Data literacy

This is the ability to read, understand, analyse and communicate data effectively. It involves interpreting data and make informed decision based on insights

Real world Example

- Refine internet my dashboard data to guide me in business strategy

⑪ Structured Data

→ Is data that is organized usually in rows and columns making it easy to store, search and analyze

Real World Example.

→ A company stores sale data (product name, price, quantity, date) in an excel table

⑫ Unstructured Data

→ This is data that is a mess of which it makes it ^{difficult} to read

Example

→ Takelab collects thousands of customer reviews, product photos and support chat messages. This doesn't follow a fixed structure like rows and columns

⑬ Data Warehouse

→ is a central place where we can have multiple databases

Example

→ A retail company stores years of its sales data from all stores in one warehouse to analyze trends.

⑭ Data Mining

→ is a process of analyzing large datasets to discover hidden patterns, trends or relationships.

Example

→ My friend's shop supermarket finds that customers who buy bread often also buy milk.

(15) Data Visualization

→ This is when you present data using visual element like charts, graphs and dashboard to make information easier to understand

Example

→ Last month I used google locker to have line graph in my dashboard showing monthly revenue trend over time

(16) Dashboard

→ This is a visual display of key data and metric usually updated in real time allowing users to monitor performance quickly

Example

→ A sales dashboard are interactive and provide a live view of performance

(17) Metric

→ This is a measurable value to see performance. but it does not make decision

Example

→ Number of people visiting in your website is a metric

(18) Key Performance Indication(KPI)

→ A KPI is a specific type of metric that is directly tied to a business objective and is used to measure success. It helps decision maker evaluate whether goal are being achieved

Example

→ An online store like Tickedot track conversion rate as a KPI to measure how many visitor become customer

19) Benchmark

→ This is a comparison that is used to compare a company's performance against industry average or competitors.

Example

A company compares its conversion rate to the industry average to evaluate performance.

20) Segmentation

→ This is a process of dividing customers into groups based on shared characteristics such as age, behavior. Unlike general analysis, segmentation focuses on grouping data to allow targeted strategies.

Example

→ A company groups customers into "high spenders" and "low spenders" to create targeted marketing campaigns.

21) Conversion Rate

→ This is a percentage of users who complete a desired action such as making a purchase or signing up.

Example

→ An online store like Amazon tracks how many website visitors actually buy a product. If 1000 people visit and 50 buy, the conversion rate is 5%.

22) Customer Acquisition Cost

→ It is the total cost a company spends to gain one new customer, including marketing and sales expenses.

Example

→ A e-commerce company spends \$10,000 on ads and gains 100 customers. The CAC is \$100 per customer.

25) Customer lifetime value

→ This is the total revenue a business expects to earn from a customer over the entire relationship. It shows how valuable a customer is in the long term.

Example

→ Subscription service like Netflix earns monthly fee from a customer over several years increasing their lifetime value.

24) Churn Rate

→ This is a percentage of customer who stop doing business with a company over a specific period. It shows how long customer a business is losing.

Example

→ A mobile network loses 50 out of 1000 customer in a month.

23) Customer Retention Rate

→ A retention rate measures the percentage of customer a business keeps over time. It shows how well a company maintains customer relationships.

Example

→ Bank keeps 900 out of 1000 customers retention rate 90%.

26) Net Promoter Score

→ Measures customer loyalty by asking how likely customer is to recommend a business to others.

Example

→ A company asks "How likely are you to recommend us?" Customer who score 9-10 are promoters while low score indicates dissatisfaction.

②7 Average order value

- It is the average amount spent each time a customer makes a purchase. It helps business understand customer spending behavior, it focuses on transaction value not customer lifetime value.

Example

- If total sales are ₹10,000 from 100 orders

②8 Sale Funnel

- Represents the stage a customer goes through from first awareness to final purchase. It shows how potential customers are converted into actual buyers.

Example

- Tracks how users move from visiting the website, adding to cart

②9 Marketing campaign

- It is a coordinated set of activities designed to promote a product, service, or brand over a specific period.

Example

- Nike launches a campaign across social media like Instagram, YouTube, and email to promote a new product line.

③0 Post-campaign Analysis

- The process of evaluating the performance of a marketing campaign after it ends. It relies on metrics like ROI to measure success.

Example

- After running ads, a company reviews performance data to see which platform generated the most sales.

③① Incremental Revenue

→ This measure the true impact of the campaign beyond normal sales. Unlike total revenue it isolate the extra income caused by the campaign and is used in ROI calculation

Example

→ If a company normally earn R50 000 but earns R70 000 during a campaign the incremental revenue is R20 000

③② Return on Investment (ROI)

This measure the profitability of an investment by comparing the profit generated to the cost of the campaign it show whether the campaign was worth the money spent

Example

→ If a campaign costs R10 000 and generate R30 000 in revenue the ROI show a strong return

③③ Return on Ad Spend

→ This measure how many is generated for every rand spent on advertising. It focuses specifically on advertising performance rather than overall profitability

Example

→ If a company spend R1000 on ads and earn R5000 in revenue the ROI is 5:1

③④ Click Through Rate

This is a percentage who click on an ad after seeing it. It measure how effective an ads is at attracting attention

Example

→ If 1000 people see an ad and 50 click it the CTR is 5%

35) Bounce Rate

→ It is a percentage of visitors who leave a website without interacting further. It indicates how engaging or relevant a webpage is.

Example

→ Users click on ads but leave immediately without viewing other pages.

36) A/B Testing

→ It is a method of comparing two versions of a marketing asset to determine which performs better. It helps businesses optimize campaigns using data. It helps metrics like CTR and conversion rate by testing variations.

Example

→ A company tests two different ad designs to see which gets more clicks.

37) Lead Generation

→ Lead generation is the process of attracting and capturing potential customers who have shown interest in a product or service. These leads can later be converted into paying customers.

Example

→ A company collects email addresses through a sign-up form on a website.

38) Marketing Attribution

→ This is the process of identifying which marketing channel or touchpoint contributed to a customer's decision to purchase. This helps businesses understand where their success is coming from. It links campaign performance to results and supports post-campaign analysis and ROI evaluation.

Example

→ A company determines whether a sale came from social media, email, or search ads.

Reference

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